

B13705023 蔡承叡 HW7

1. PVE

1.

- `mkdir local_libvirt`
- `mkdir images` 進到自己的 local 存 disk 的地方，因為沒 `sudo` 不能動 `/var`
- `qemu-img create -f qcow2 b13705023-1.qcow2 10G`

```
virt-install
--name b13705023-1
--vcpus 2
--memory 8192
--disk path=~/.local_libvirt/images/b13705023-1.qcow2,size=10,format=qcow2
--cdrom /tmp2/hw7/proxmox.iso
--network bridge=br0,mac=52:54:70:50:23:01
--graphics vnc,listen=0.0.0.0
--os-variant debian11
--noautoconsole
```

- `virsh --connect qemu:///system list --all` 可以檢查自己虛擬機狀態
- `virsh vncdisplay b13705023-1` 出現 `:1` 表示開在 5901 端口
- 去 vnc viewer 輸入 140.112.187.51:5901 連上



Welcome to Proxmox Virtual Environment

- - Install Proxmox VE (Graphical)
 - Install Proxmox VE (Terminal UI)
 - Install Proxmox VE (Terminal UI, Serial Console)
 - Advanced Options

enter: select, arrow keys: navigate, e: edit entry, esc: back

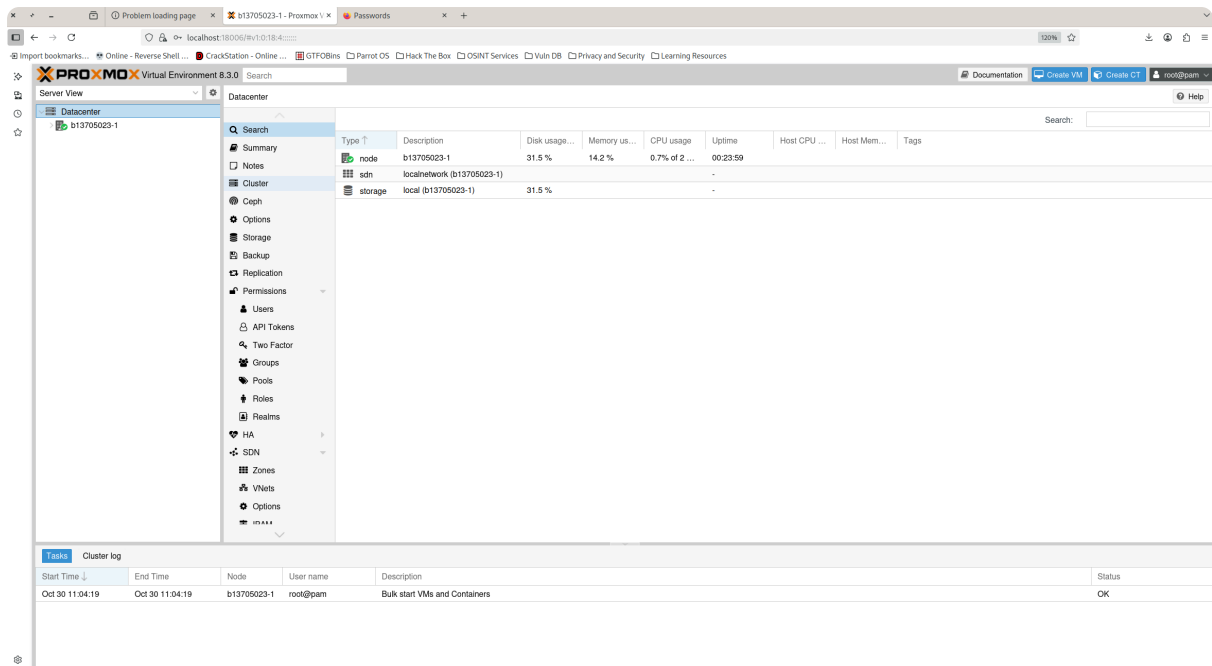
- 幾乎都是原始設定就好，記得檢查 hostname 和檔案系統是不是 ext4

2.

```
b13705023@nasaws1:~/local_libvirt/images$ ping 192.168.141.200
PING 192.168.141.200 (192.168.141.200) 56(84) bytes of data.
64 bytes from 192.168.141.200: icmp_seq=1 ttl=64 time=0.144 ms
64 bytes from 192.168.141.200: icmp_seq=2 ttl=64 time=0.674 ms
64 bytes from 192.168.141.200: icmp_seq=3 ttl=64 time=0.277 ms
```

3.

- `ssh -L 18006:192.168.141.200:8006 b13705023@140.112.187.51` 保持連線開啟
- 連上 <https://localhost:18006>
- 帳號：root 密碼：剛剛 vnc 裡面設的



參考資料：

<https://blog.gtwang.org/linux/kvm-qemu-virt-install-command-tutorial/>

<https://ask.csdn.net/questions/8436099>

<https://ithelp.ithome.com.tw/articles/10265378>

2. Create VM

1.

- `qemu-img create -f qcow2 b13705023-1-zfs.qcow2 20G`
- `virsh attach-disk b13705023-1 ~/local_libvirt/images/b13705023-1.qcow2 vdb --driver qemu --subdriver qcow2 --targetbus virtio --persistent` 把剛剛的 qcow2 裝到 PVE 上
- `ssh root@192.168.141.200` 進入 VM
- `zpool create -f tank /dev/vdb` 建立 zfs pool on /dev/vdb, name = tank
- `zfs create tank/vmdata`

參考資料：

<https://www.youstable.com/blog/use-zfs-on-linux/>

2.

- `virsh edit b13705023-1`

- 選用 nano 之後在 <devices> 裡面新增

```
<interface type='user'>
  <mac address='52:54:90:21:23:02' />
  <model type='virtio' />
</interface>
```

- virsh shutdown b13705023-1
- virsh start b13705023-1 (這邊用 reboot 不知道為啥不行 反正不要 reboot 就對了網卡會出不來)
- 連上 PVE 之後 ip a 看一下新網卡叫啥
- 進 /etc/network/interfaces 加入

```
auto enp8s0
iface enp8s0 inet static
    address 10.0.2.15
    netmask 255.255.255.0
    gateway 10.0.2.2
    dns-nameservers 8.8.8.8
```

- systemctl restart networking

參考資料：

claude

3.

- /etc/network/interfaces 加入

```
auto vmbr1
iface vmbr1 inet static
    address 10.0.2.15
    netmask 255.255.255.0
    gateway 10.0.2.2
    bridge-ports enp8s0
    bridge-stp off
    bridge-fd 0
    dns-nameservers 8.8.8.8
```

- ifdown enp8s0
- ifup vmbr1

- 等一下之後 `ip a` 就看的到網卡抓到 ip 了，但可能 `apt update` 還是沒辦法，表示還是沒網路，那可以試試看去修改 `/etc/resolv.conf` 看看是不是 DNS 問題，把裡面改成

```
nameserver 8.8.8.8
nameserver 1.1.1.1
```

```
root@b13705023-1:~# apt update
Get:1 http://ftp.tw.debian.org/debian bookworm InRelease [151 kB]
Get:2 http://ftp.tw.debian.org/debian bookworm-updates InRelease [55.4 kB]
Get:3 http://security.debian.org bookworm-security InRelease [48.0 kB]
Get:4 http://ftp.tw.debian.org/debian bookworm/main amd64 Packages [8791 kB]
Get:5 http://ftp.tw.debian.org/debian bookworm/main Translation-en [6109 kB]
Get:6 http://ftp.tw.debian.org/debian bookworm/contrib amd64 Packages [53.5 kB]
Get:7 http://ftp.tw.debian.org/debian bookworm/contrib Translation-en [48.4 kB]
Get:8 http://ftp.tw.debian.org/debian bookworm-updates/main amd64 Packages [692
B]
Get:9 http://ftp.tw.debian.org/debian bookworm-updates/main Translation-en [544
B]
Get:10 http://security.debian.org bookworm-security/main amd64 Packages [284 kB]
Get:11 http://security.debian.org bookworm-security/main Translation-en [172 kB]
Get:12 http://security.debian.org bookworm-security/contrib amd64 Packages [896
B]
```

參考資料：

claude

<https://unix.stackexchange.com/questions/671992/how-to-add-an-additional-network-interface-on-a-kvm-vm>

4.

- `scp /tmp2/hw7/alpine.iso root@192.168.141.200:/var/lib/vz/template/iso/` 把 alpine 的 iso 抓到 PVE 裡面
- 去 web gui -> Datacenter -> Storage -> Add -> ZFS -> 把剛剛在 cli 創的加進去
- 上面的 Create VM, 然後照這個配置去配 (disks 部分要選 tank)

Key ↑	Value
cores	1
cpu	x86-64-v2-AES
ide2	local:iso/alpine.iso,media=cdrom
memory	1024
name	alpine-b13705023
net0	virtio,bridge=vbr1,firewall=1
nodename	b13705023-1
numa	0
ostype	l26
scsi0	tank:6,iothread=on
scsihw	virtio-scsi-single
sockets	1
vmid	100

- 左邊應該會出現那個 alpine 機器，點進去 Console 直接 power on
- root 登入
- setup-alpine
- Keyboard: us 兩次
- Hostname: alpine-b13705023
- Network interface: eth0
- IP: dhcp 讓他自己抓
- Timezone: Asia/ Taipei
- Proxy: none
- APK Mirror: f 讓他自己找
- User: no (第 5 題建立)
- SSH: openssh
- Disk: sda
- How would...: sys
- 以此類推...
- reboot

參考資料：

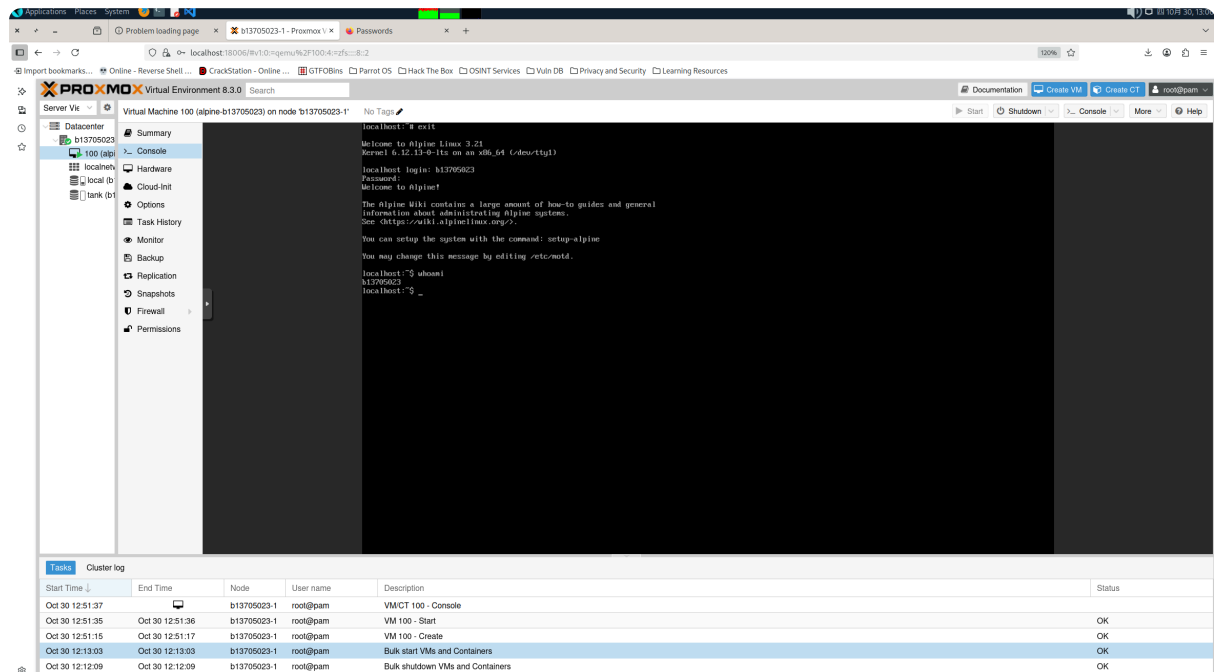
<https://hackmd.io/@joshhu/rJjG5guLa>

5.

- root login

- adduser b13705023
- 設定密碼
- exit
- 用 b13705023 登入

6.



Cluster and HA

1.

2.

- qemu-img create -f qcow2 /var/lib/libvirt/images/b13705023-2.qcow2 10G
- qemu-img create -f qcow2 /var/lib/libvirt/images/b13705023-2-disk2.qcow2 20G

```
virt-install
  --name b13705023-2
  --vcpus 2
  --memory 8192
  --disk path=~/.local/libvirt/images/b13705023-2.qcow2,size=10,format=qcow2,bus
  --disk path=~/.local/libvirt/images/b13705023-2-disk2.qcow2,size=20,format=qcow2,bus
  --network bridge=br0,mac=52:54:70:50:23:03,model=virtio
  --network user,mac=52:54:70:50:23:04,model=virtio
  --cdrom /tmp2/hw7/proxmox.iso
  --graphics vnc,listen=0.0.0.0
  --os-variant debian11
  --noautoconsole
```

- `virsh vncdisplay b13705023-2` 出現 :3 , 所以開 vnc 連
- 進去調整設定 , 幾乎跟第一題一樣 , 不過要確定 IP 設在 br0 的網段裡面 , gateway 用 br0 的 ip , DNS 可以先調 8.8.8.8 等等就不用去 `/etc/resolv.conf` 調整
- 設定好之後 `ssh root@192.168.141.201` 用剛剛的密碼登入
- `zpool create -f tank /dev/vdb`
- `zfs create tank/vmdata`
- 上面就是做跟 pve1 一樣的事情 , 把 vdb 拿去做 zfs
- `nano /etc/network/interfaces`
- 加入

```
auto enp2s0
iface enp2s0 inet dhcp (可和 PVE1 一樣手動設或是用 DHCP)
```

```
auto vmbr1
iface vmbr1 inet static
    address 10.0.2.15
    netmask 255.255.255.0
    gateway 10.0.2.2
    bridge-ports enp2s0
    bridge-stp off
    bridge-fd 0
    dns-nameservers 8.8.8.8
```

- `systemctl restart networking`
- `ip a` 應該能看到網卡拿到 IP , 然後 `apt update` 應該要能 `get` 表示有網路


```

root@b13705023-2:~# apt update
Get:1 http://ftp.tw.debian.org/debian bookworm InRelease [151 kB]
Get:2 http://ftp.tw.debian.org/debian bookworm-updates InRelease [55.4 kB]
Get:3 http://security.debian.org bookworm-security InRelease [48.0 kB]
Get:4 http://ftp.tw.debian.org/debian bookworm/main amd64 Packages [8791 kB]
Get:5 http://ftp.tw.debian.org/debian bookworm/main Translation-en [6109 kB]
Get:6 http://security.debian.org bookworm-security/main amd64 Packages [284 kB]
Get:7 http://ftp.tw.debian.org/debian bookworm/contrib amd64 Packages [53.5 kB]
Get:8 http://ftp.tw.debian.org/debian bookworm/contrib Translation-en [48.4 kB]
Get:9 http://ftp.tw.debian.org/debian bookworm-updates/main amd64 Packages [6924 B]
Get:10 http://ftp.tw.debian.org/debian bookworm-updates/main Translation-en [544 B]

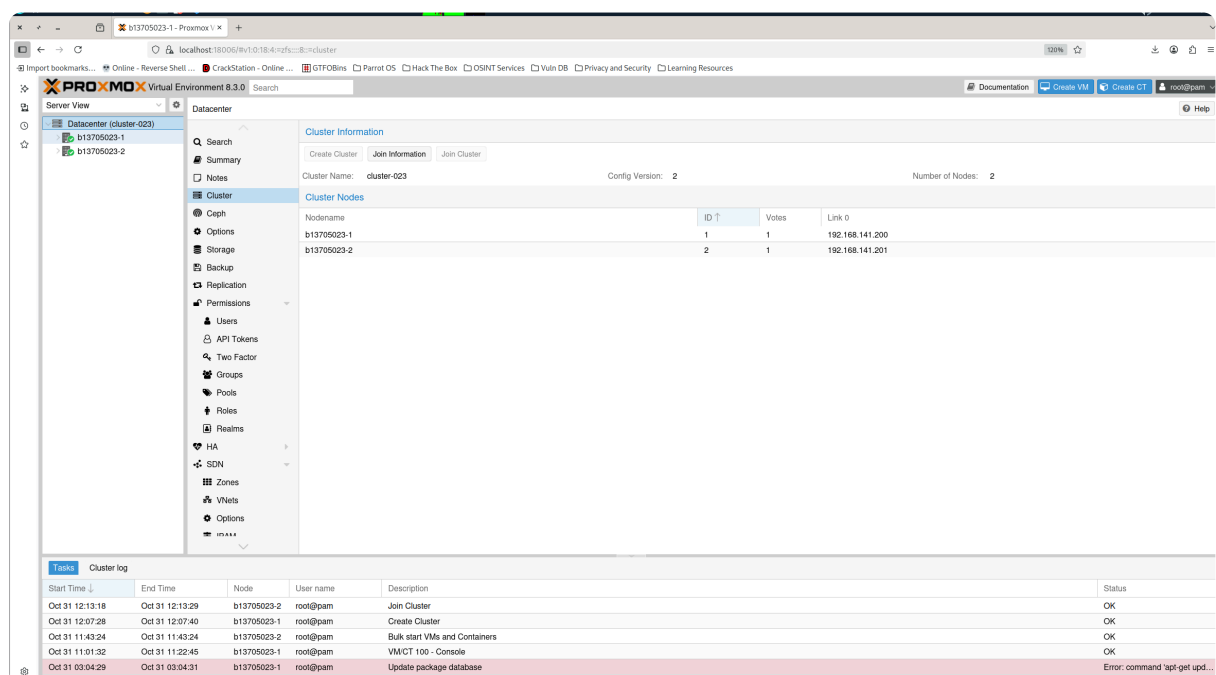
```

參考資料：

Part 1 and 2

3.

- 在 pve1 的 webgui 裡面 -> Datacenter -> Cluster -> Create Cluster -> 命名，link 0 ip 192.168.141.200 -> Create
- 去剛剛創建的 Cluster -> Join Information -> Copy Information
- 再創一個 terminal，模仿 part 1 做 ssh 序列埠轉送
- ssh -L 28006:192.168.141.201:8006 b13705023@140.112.187.51
- 就可以連上 <https://localhost:28006>
- 去 Cluster -> Join Cluster -> 貼上剛剛複製的一大串 json -> Password 輸入 pve1 的 root 密碼 -> Join



參考資料：

Part 1

<https://bachelor-tech.com/how-to-join-two-proxmox-nodes-into-a-cluster-pve-8-x/>

https://www.reddit.com/r/Proxmox/comments/112bl8i/joining_cluster_connection_error/

4.

- Shutdown 在跑的 alpine
- 因為 b13705023-2 已經被加進來了，所以可以在 pve1 的 webgui 調整 pve2 的設定，所以先去 Datacenter -> Storage -> Add -> ZFS -> 設 ID，Nodes 換成 b13705023-2 -> Add
- 把 b13705023-1 的 tank 改成 Nodes: no restrictions, 這樣才能 migrate
- 還要在 alpine 的 Hardware 設定裡面選 CD/DVD Drive -> Edit -> Do not use any media
- 右鍵機器 -> Migrate

Task viewer: VM 100 - Migrate (b13705023-1 ----> b13705023-2)

Output Status

Stop Download

```
2025-10-31 12:39:56 starting migration of VM 100 to node 'b13705023-2' (192.168.141.201)
2025-10-31 12:39:56 found local disk 'tank:vm-100-disk-0' (attached)
2025-10-31 12:39:56 copying local disk images
2025-10-31 12:39:56 full send of tank:vm-100-disk-0@__migration__ estimated size is 6.61K
2025-10-31 12:39:56 total estimated size is 6.61K
2025-10-31 12:39:56 TIME SENT SNAPSHOT tank:vm-100-disk-0@__migration__
2025-10-31 12:39:56 successfully imported 'tank:vm-100-disk-0'
2025-10-31 12:39:57 volume 'tank:vm-100-disk-0' is 'tank:vm-100-disk-0' on the target
2025-10-31 12:39:57 migration finished successfully (duration 00:00:02)
TASK OK
```

PROXMOX Virtual Environment 8.3.0 Search Documentation Create VM Create CT root@pam

Server View

Datacenter (cluster-023)

- b13705023-1
 - localnetwork (b13705023-1)
 - local (b13705023-1)
 - tank (b13705023-1)
- b13705023-2
 - 100 (alpine-b13705023)
 - localnetwork (b13705023-2)
 - local (b13705023-2)
 - tank (b13705023-2)

Virtual Machine 100 (alpine-b13705023) on node 'b13705023-2' No Tags Start Shutdown

Summary

Console Hardware Cloud-Init Options Task History Monitor Backup Replication Snapshots Firewall Permissions

Hardware

Memory	1.00 GiB
Processors	1 (1 sockets, 1 cores) [x86-64-v2-AES]
BIOS	Default (SeaBIOS)
Display	Default
Machine	Default (i440fx)
SCSI Controller	VirtIO SCSI single
CD/DVD Drive (ide2)	none,media=cdrom
Hard Disk (scsi0)	tank:vm-100-disk-0,iouthread=1,size=6G
Network Device (net0)	virtio=BC:24:11:DB:86:A3,bridge=vbr1,firewall=1

參考資料：

<https://forum.proxmox.com/threads/proxmox-migration-non-cluster.156175/>
claude

5.

- Datacenter -> Replication -> Add -> VM ID: 100, Target: b13705023-1, Schedule: */30 -> Create

The screenshot shows the Proxmox VE 8.3.0 interface. On the left, the 'Datacenter (cluster-023)' is expanded, showing two nodes: 'b13705023-1' and 'b13705023-2'. Each node has associated resources like 'localnetwork', 'local', and 'tank'. The 'Replication' menu item is selected in the left sidebar. The main panel displays the 'Replication' configuration page with a table showing one replication job.

Enabled	Guest ↑	Job ↑	Target	Sched...	Comment
✓	100	0	b13705023-1	*/30	

6.

- Datacenter -> HA -> Groups -> Create -> ID: group-b13705023, Node 全選 其他預設 -> Create

The screenshot shows the Proxmox VE 8.3.0 interface. On the left, the 'Datacenter (cluster-023)' is expanded, showing two nodes: 'b13705023-1' and 'b13705023-2'. Each node has associated resources like 'localnetwork', 'local', and 'tank'. The 'HA' menu item is selected in the left sidebar, and the 'Groups' sub-menu is also selected. The main panel displays the 'HA Groups' configuration page with a table showing one group.

Group ↑	restricted	nofailback	Nodes
group-b13705023	No	No	b13705023-2,b13705023-1

7.

- Datacenter -> HA -> Add -> VM: 100, Group: group-b13705023 其他預設 -> Add

PROXMOX Virtual Environment 8.3.0

Server View

Datacenter (cluster-023)

b13705023-1

localnetwork (b13705023-1)

local (b13705023-1)

tank (b13705023-1)

b13705023-2

100 (alpine-b13705023)

localnetwork (b13705023-2)

local (b13705023-2)

tank (b13705023-2)

Permissions

Users

API Tokens

Two Factor

Groups

Pools

Roles

Realms

HA

Groups

Fencing

SDN

Zones

VNets

Status

Type	Status
quorum	OK
master	b13705023-1 (active, Fri Oct 31 13:12:10 2025)
lrm	b13705023-1 (idle, Fri Oct 31 13:12:13 2025)
lrm	b13705023-2 (active, Fri Oct 31 13:12:11 2025)

Resources

Add Edit Remove

ID	State	Node	Name	Max. Restart	Max. Reloc...	Group
vm:100	started	b13705023-2	alpine-b13...	1	1	group-b13705023